

17378

STATE OF MICHIGAN
DEPARTMENT OF CONSERVATION
SUPERVISOR OF WELLS

APPLICATION FOR PERMIT TO DRILL A WELL FOR OIL, GAS, GEOLOGICAL TEST,
BRINE DISPOSAL, OR KEY WELL FOR SECONDARY RECOVERY

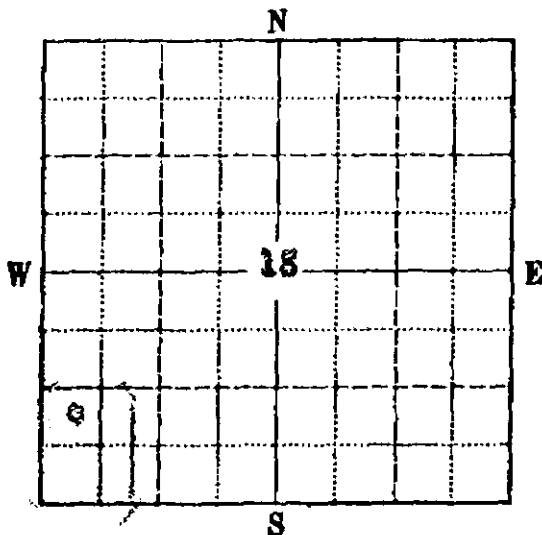
Under Authority of Act 61, P. A. 1939 and Act 326, P. A. 1937

See Reverse Side Before Filling Out This Application

Department of Conservation
Geological Survey Division
Lansing 13, Michigan
Gentlemen:

Allegan, Michigan January 25, 1952

Application is hereby made for permit to drill a well for Oil



(Locate well and lease accurately on
this section plat—640 acres)

Acres in lease 30

Locate well in two directions from nearest lines of quarter section and property ownership lines:—

990 ft. from ~~North~~ south line of quarter section, 330 ft. from ~~East~~ west line of quarter section;

330 ft. from north ~~East~~ property line, 330 ft. from ~~West~~ west property line.

Number of feet from nearest producing well 660 Kind of tools to be used cable

Well is to be completed in Traverse formation. Intended depth 1375'

Program of casing, sealing, and completing well Drift, Coldwater, Antrim and Traverse. Drive 8" pipe through the drift. Set 6-5/8" casing on a cave or water string if necessary. Set 5-3/16" casing on top of the Traverse. All oil, gas and/or water encountered to be confined to the formation in which it occurs.

Well location is in Cleared farm land
Cleared farm land. Open grass land. Brush land. Timbered land.

Are there any lakes, rivers, streams, ditches, drainage-ways, swamps or marshes nearer than 330 feet from well location?
no If yes, describe and give distance from location and specify precautions to be taken to protect fresh water from well wastes

Are there any buildings or other structures within 75 feet of the proposed location? no
Yes—No

Applicant agrees to comply with the provisions of Act 61, Public Acts of 1939, and Act 326, Public Acts of 1937, and rules and regulations pertaining to the locating, drilling, casing, sealing, completing, producing, and plugging of wells.

Applicant will notify the nearest field office of the department when ready to commence drilling operations.

Address correspondence and permit to Louis Zellman, P. O. Box 114, Allegan, Michigan

Phone 407

bea

Signed LOUIS ZELLMAN
Owner of lease rights

By Louis Zellman
Authorized representative

PERMITS MAY BE TRANSFERRED ONLY WITH PERMISSION OF THE SUPERVISOR OF WELLS

P#17373

L. & M. Steinmiller #1

STATE OF MICHIGAN
DEPARTMENT OF CONSERVATION
SUPERVISOR OF WELLS

PERMIT TO DRILL A WELL FOR OIL OR GAS

(POST CLOTH PERMIT AT WELL LOCATION)

Permit No. 17378

Date Issued January 20, 1952

Permission is hereby granted to Louie Vollman
Owner

P. O. Box 114, Allegan, Michigan
Address

Lease Name Louie & Margaret Steinmiller

Well Number 1

To drill a well for Oil

in the Wet and Dry, Section 18, T.1N., R.15W., Allegan County, Lee Township,

and located 990' from south and 330' from west line of quarter section;

and located 330' from north and 330' from west property line.

This permit is issued subject to the provisions and requirements of Act 61, P. A. 1939 and Act No. 326, P. A. 1937, as amended, and rules, regulations, requirements, or orders issued by the Supervisor of Wells and the Department of Conservation.

GERALD E. EDDY, SUPERVISOR OF WELLS

By

Lyle W. Rice

CAUTION

DRILLING OPERATIONS NOT TO BE COMMENCED until all the well location requirements have been completed and the permit has been posted in a conspicuous place at the well site.

WASTE IS PROHIBITED in the drilling, producing, and plugging of this well. This includes underground, surface, and market waste.

DAMAGE TO FRESH WATERS, both underground and surface, is prohibited in the drilling, producing, and plugging of this well.

UNNECESSARY DAMAGE TO LIFE AND PROPERTY is prohibited in the drilling, producing, and plugging of this well.

Department offices at: Lansing, Phone 5-7185
Mount Pleasant, Phone 3-4591
Cadillac, Phone 138
Allegan, Phone 89
Grand Rapids, Phone 8-3007

PERMITS MAY BE TRANSFERRED ONLY WITH PERMISSION OF THE SUPERVISOR OF WELLS



STATE OF MICHIGAN
DEPARTMENT OF CONSERVATION

LOG OF OIL, GAS OR TEST WELL

TO BE FILED WITH SUPERVISOR OF WELLS WITHIN 30 DAYS
AFTER COMPLETION OF WELL (ACT 81 P. A. 1939
AND ACT 326 P. A. 1937)

LOCATE WELL ACCURATELY

Permit No. 17378Owner or operator Louis ZellmanAddress P. O. Box 114, Allegan, MichiganWell No. 1, Farm Louis & Margaret SteinmillerLocation NW SW SW Sec. 18 Twp. 1N Range 15WTownship Lee County AlleganFootage from line of quarter Sec. NL SL 980 EL WL 350Type of well Dry Hole Elevation 636.2'
Oil, Gas, or Dry Hole From Derrick FloorName of producing formation _____ Top of producing formation _____ Total depth 1200Date drilling began February 13, 1952 Date drilling completed February 23, 1952Drilling contractor W. E. Wyse Address 231 North St., Allegan, Michigan

WATER ZONES

Name	From	To	Name	From	To
<u>Traverse</u>	<u>1200</u>				

OIL OR GAS PAYS

Name	From	To	Content	Name	From	To	Content

CASING RUN

CASING PULLED

Size	Weight per Foot	Amount Set	Kind of Shoe	Size	Amount Pulled
<u>8"</u>		<u>133'</u>			
<u>6-5/8" Reduced Hole</u>		<u>133'</u>			
<u>5-3/16"</u>		<u>1162'</u>			

MUDDING OR CEMENTING RECORD

ACID OR SHOOTING RECORD

Size Csg.	Where Set	Number Sacks	Method Used	Date of Shot or Acid Treat	From	To	Number Quarts Used or Number Gallons Acid

Rotary tools from _____ feet to _____ feet. Cable tools from 0 feet to 1200 feet.

Natural initial production first 24 hours _____ Bbls. After acid or shot _____ Bbls.

If gas well, cu. ft. per 24 hours _____ Rock pressure, lbs. per sq. in. _____

The above information is complete and correct.

Date February 25, 1952

Signed

Title

OPERATOR

FEB 26 1952

FORMATION RECORD

FROM	TO	FORMATION	FROM	TO	FORMATION
0	135	Drift			
133	200	Gray Shale			
200	228	Dark Lime			
228	420	Gray Shale			
420	460	Coldwater Limestone			
460	517	Gray Shale			
517	537	Red Rock			
537	960	Green Shale			
960	1020	Light Brown Shale			
1020	1135	Dark Brown Shale			
1135	1180	Gray Shale & Lime Shells			
1180		Top of Traverse			
	1200	TOTAL DEPTH			
1200		300 feet water/ hour			

Date of Completion 2-23-52
Date Last Produced DRY HOLE
Last Production oil-gas-water
Date work is to start will not fy

STATE OF MICHIGAN
DEPARTMENT OF CONSERVATION
SUPERVISOR OF WELLS

Leave Blank
Permit No. 17378
Date Issued _____
D.P. No. _____
Date Issued _____

APPLICATION FOR PERMISSION TO:
(Submit in Quadruplicate)

~~REPAIR~~ - ~~REWORK~~ - ABANDON
(Cross out those which do not apply)

Supervisor of Wells
Department of Conservation
Lansing 13, Michigan

Allegan Michigan February 25 19 52

In compliance with the provisions of Act 61, Public Acts of 1939, and Act 326, Public Acts 1937 and the rules and regulations issued and adopted thereunder permission is requested to ~~Reopen~~ - ~~Re-work~~ - Abandon the

Louis Zellman Louis & Margaret Steinmiller # 1

Well Owner

Well Name and Number

NW $\frac{1}{4}$, SW $\frac{1}{4}$, SW $\frac{1}{4}$, Sec. 18 T. 1N N.S., R. 15W E.W.

Lee

Allegan

Township

County

The work to be done by: W. E. Wyse
Operator or Contractor

Outline the present condition of the well giving casing, cementing, formation, and production data.

8" 133'	Cw. Ls. 420-60	Tr. Ls. 1180
6-5/8" 133' RH	Cw. RR. 517-37	300' Wtr./ hr. 1200
5-3/16" 1162' RH	Lt. Ant. 960	
	Dk. Ant. 1020	T. D. 1200
	Tr. Fm. 1135	

Describe the proposed work in detail.

Fill the bottom hole from the total depth to 730' using approximately 205 pails of mud.

Bridge solidly at 133' and fill to the 3" drive pipe using 25 pails of mud.

Start the 3" drive pipe and fill to the surface with pit mud, as the drive is being pulled.
If the drive pipe is to remain in the hole temporarily, cover over safely at the surface.

CALL THE DEPARTMENT OF CONSERVATION AT ALLEGAN (PHONE 89) BEFORE 9:30 A.M. WHEN READY TO START PLUGGING OPERATIONS AND WHEN READY TO START DRIVE PIPE.

Plugging equipment on the location shall include a line of sufficient length to run to the total depth and tools of sufficient size and weight to assure the proper cleaning out of obstructions and setting of required bridges.

Field Office ALLEGAN
Approved [Signature]
Date 2-25-52

LOUIS ZELLMAN

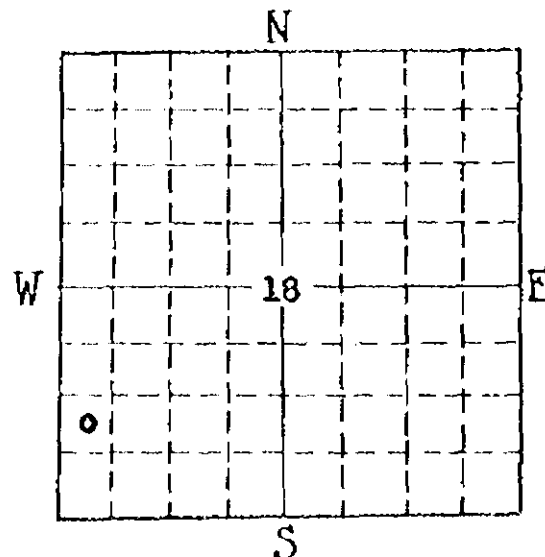
Name of Company or Operator

Authorized Representative

Record of proposed work to be filed within 30 days after its completion.

bjb

G.S. 37 6-6-44



STATE OF MICHIGAN
DEPARTMENT OF CONSERVATION

WELL PLUGGING RECORD

TO BE FILED WITHIN THIRTY DAYS AFTER FINAL ABANDONMENT

Permit No. 17378 Date February 25 1952

Owner or operator Louis Zellman Address P. O. Box 114, Allegan, Michigan

Well No. 1, Farm Steinmiller, Sec. 18, T. 1N., N. S., R. 15W., E. W. NW 1/4 SW 1/4 SW 1/4

Township Lee County Allegan

Located 990 feet from the $\left\{ \begin{smallmatrix} N \\ S \end{smallmatrix} \right\}$ line of quarter Sec. and 330 feet from the $\left\{ \begin{smallmatrix} E \\ W \end{smallmatrix} \right\}$ line of quarter Sec.

Type of well Dry Hole Name of producing sand
(Oil, Gas, or Dry Hole)

Top sand Bottom sand Total depth 1200

Date plugging began February 23, 1952 Date plugging completed February 23, 1952

Was permission of Department of Conservation secured before plugging began? Yes

Name of representative of Department who supervised plugging B. C. Ackerman

OIL, GAS AND WATER SANDS

CASING RECORD

ACKERMAN

FIELD ADMINISTRATION
APPROVAL FOR CLEAN-UP

DATE 9-23-52

BY [Signature]

Describe in detail how well was plugged Filled the bottom hole from the total depth to 780' using approximately 205 pails of mud. Bridged solidly at 183' and filled to the 8" drive pipe using 25 pails of mud. The 8" drive pipe will remain in the hole for pulling contractors. Hole is covered over safely at the surface. (8" drive pipe (all- 133') pulled by the Michigan Pipe Company on 5-16-52 and mudded to the surface. Well surface ok on 6-19-52. BCA)

Names of workmen present when well was plugged W. E. Wyse & Al Santee

Signed LOUIS ZELLMAN By Louis Zellman

Address P. O. Box 114, Allegan, Michigan OPERATOR
Title

State of Michigan, County of Allegan ss:

I, Louis Zellman, do hereby swear that the above statements are true and correct to the best of my knowledge and belief.

Subscribed and sworn to before me this 26th day of February 1952

My commission expires 2-4-53 Notary Public

O. K. by Department representative B. C. Ackerman

SEP 23 1952

4RA

Co. Louis Zellman
 Farm Louis & Margaret Steinmiller
 Twp. Lee Co. Allegan
 Loc. NW SW SW Sec. 13 T. 1N R. 15W
 990' from S and 330' from W line of 1/4 Sec.
 330' from N and 330' from W property line
 Contractor W. E. Wyse
 Comm. 2-13-52 Comp. 2-23-52
 2-25-52

CASING RECORD

				Size	Length	Pulled
				14"		
				10"		
	18			8 1/4"	133	
0				6 1/4"	133 RH	
				5 1/4"	1162 RH	

MUDDING AND CEMENTING RECORD

Date	Size	No. Sacks	Company
	10"		
	8 1/4"		
	6 1/2"		
	5 1/4"		

GEOLOGICAL FORMATION RECORD

Elevation	636.2'	
Drift to		
Cw. Ls.	420-60	
Cw. RR.	517-37	
Lt. Ant.	960	
Dk. Ant.	1020	
Tr. fm.	1135	
Tr. Ls.	1180	
300' wtr./hr.	1200	
		T. D. 1200
		P. B.
		I. P. (N) D & A
		I. P. (A)
		Plug. Comp.

Fill to 730' using 205 mud.

Bridge at 133' and fill to 3" using 25 mud.

2-25-52 B.H. mudded and 5"
pulled on 2-23-52. Not present
LRF

3-14-52 d.p. still in hole. ~~OK~~

5-16-52 Michigan Pipe Co. pulled
all (132') of 8" d.p. & filled
to surface. Mousing off ~~OK~~

6-19-52 Well surface ok.
Pit not covered. ~~OK~~



FIELD BOREHOLE LOG

BOREHOLE NO: **120TH-B001**TOTAL DEPTH: **50 FT**

PROJECT INFORMATION						DRILLING INFORMATION			
PROJECT:		120TH Ave PFAS Study Area				CONTRACTOR:		Mateco Drilling Company	
SITE LOCATION:		Robinson Township, MI				CREW CHIEF:		Mitch Slachter	
PROJECT NO.:		60598574				DRILL RIG TYPE:		Geoprobe 7822	
PROJECT MANAGER:		Matt Vander Eide				DRILLING METHOD:		Direct Push	
LOGGED BY:		Kaitlyn Eicholtz				HOLE DIAMETER:		2 1/4"	
CREATED BY:		Kaitlyn Eicholtz				DATE START:		2/25/19 08:50	
						DATE END:		2/25/19 15:45	

DEPTH	SAMPLE TYPE	ATTEMPT	RECOVERY	SOIL SYMBOLS	USCS	SOIL DESCRIPTION	WELL CONSTRUCTION	SAMPLE
0	HAND AUGER				OL	OL: 10YR2/2 POORLY SORTED SILTY FINE SAND, non plastic, sub rounded sand grains and sub rounded to sub angular gravel, 40% silt, 30% fine sand, 20% medium sand, 5% coarse sand, 5% fine gravel, organic silt, Topsoil, 0ft to 0.7ft bgs is frozen, moist	0	
					SM	SM: 10YR4/6 POORLY SORTED SILTY FINE TO MEDIUM SAND WITH GRAVEL, non plastic, sub rounded to sub angular sand grains and gravel, 20% silt, 20% fine sand, 20% medium sand, 10% coarse sand, 20% fine gravel, 10% coarse gravel, gravel base, moist	-1	
					SP-SM	SP-SM: 10YR6/8 WELL SORTED FINE SAND WITH SILT, non plastic, sub rounded to sub angular sand grains and gravel, 10% silt, 80% fine sand, 10% medium sand, trace coarse sand, trace fine gravel, massive, loose, moist	-2	
					SP	SP: 10YR6/8 WELL SORTED FINE SAND, non plastic, sub rounded to sub angular sand grains and gravel, 5% silt, 85% fine sand, 10% medium sand, trace coarse sand, trace fine gravel, iron staining, massive, loose, wet	-3	
5	DIRECT PUSH				SP	SP: 10YR6/8 WELL SORTED FINE SAND, non plastic, sub rounded to sub angular sand grains, 5% silt, 85% fine sand, 10% medium sand, trace coarse sand, iron staining, massive, loose, wet	-4	MW Sample: 120TH-B001-8 GW190308 1025RL
					SP	SP: 10YR5/6 WELL SORTED FINE SAND, non plastic, sub rounded to sub angular sand grains, 5% silt, 90% fine sand, 5% medium sand, trace coarse sand, massive, medium dense, wet	-5	
						No Recovery	-6	
10							-7	
							-8	
							-9	
							-10	

FIELD BOREHOLE LOG

BOREHOLE NO: 120TH-B001

TOTAL DEPTH: 50 FT

DEPTH	SAMPLE TYPE	ATTEMPT	RECOVERY	SOIL SYMBOLS	USCS	SOIL DESCRIPTION	WELL CONSTRUCTION	SAMPLE
	DIRECT PUSH				SP	SP: 10YR5/4 WELL SORTED FINE SAND, non plastic, sub rounded to sub angular sand grains, 5% silt, 80% fine sand, 10% medium sand, 5% coarse sand, medium dense, wet, coarsening downward		
					SP	SP: 10YR5/4 WELL SORTED FINE SAND, non plastic, sub rounded to sub angular sand grains, 5% silt, 75% fine sand, 15% medium sand, 5% coarse sand, medium dense, wet, coarsening downward	-11	
					SP	SP: 10YR5/4 WELL SORTED FINE SAND, non plastic, sub rounded to sub angular sand grains and gravel, 5% silt, 55% fine sand, 20% medium sand, 10% coarse sand, trace fine gravel, medium dense, wet, coarsening downward	-12	
					SP	SP: 10YR5/4 WELL SORTED FINE SAND, non plastic, sub rounded to sub angular sand grains and gravel, 5% silt, 50% fine sand, 30% medium sand, 15% coarse sand, trace fine gravel, medium dense, wet, coarsening downward		
					SP	SP: 10YR5/4 WELL SORTED FINE SAND, non plastic, sub rounded to sub angular sand grains, 5% silt, 80% fine sand, 10% medium sand, 5% coarse sand, medium dense, wet, fining downward	-13	
						No Recovery	-14	
15					SW	SW: 10YR5/4 POORLY SORTED FINE SAND, non plastic, sub rounded to sub angular sand grains and gravel, 5% silt, 60% fine sand, 30% medium sand, 5% coarse sand, trace fine gravel, medium dense, wet, sharp boundary below	-15	
	DIRECT PUSH				SP	SP: 10YR5/4 WELL SORTED FINE SAND, non plastic, sub rounded to sub angular sand grains, 5% silt, 90% fine sand, 5% medium sand, trace coarse sand, massive, medium dense, wet, sharp boundary below	-16	
					SW	SW: 10YR5/4 POORLY SORTED FINE SAND WITH GRAVEL, non plastic, sub rounded to sub angular sand grains and gravel, 5% silt, 45% fine sand, 15% medium sand, 20% coarse sand, 10% fine gravel, 5% coarse gravel, gravel average 5mm to 28mm massive, medium dense, wet, sharp boundary below	-17	
						No Recovery	-18	
							-19	
20	DIRECT PUSH				SW	SW: 10YR5/4 POORLY SORTED FINE SAND WITH GRAVEL, non plastic, sub rounded to sub angular sand grains and gravel, 5% silt, 45% fine sand, 15% medium sand, 20% coarse sand, 10% fine gravel, 5% coarse gravel, gravel average 5mm to 26mm massive, medium dense, wet, fine sand lenses (20.1ft to 20.3ft bgs and 22.1ft to 22.3ft bgs)	-20	
							-21	
							-22	

FIELD BOREHOLE LOG						BOREHOLE NO:	120TH-B001	
						TOTAL DEPTH:	50 FT	
DEPTH	SAMPLE TYPE	ATTEMPT	RECOVERY	SOIL SYMBOLS	USCS	SOIL DESCRIPTION	WELL CONSTRUCTION	SAMPLE
25	DIRECT PUSH					No Recovery		VAS Collection: 120TH-B001-25 GW190225 1135KME
						Lithology not recovered due to VAS Collection		
	DIRECT PUSH				SW	SW: 10YR5/3 POORLY SORTED FINE SAND WITH GRAVEL, non plastic, sub rounded to sub angular sand grains and gravel, trace silt, 35% fine sand, 20% medium sand, 25% coarse sand, 15% fine gravel, 5% coarse gravel, gravel average 5mm to 30mm, massive, medium dense, wet, with occasional fine sand lenses (5cm to 10cm in thickness), with occasional coarse sand lenses (1cm to 3cm in thickness)		
						SP: 10YR5/3 WELL SORTED FINE SAND, non plastic, sub rounded to sub angular sand grains and gravel, trace silt, 75% fine sand, 20% medium sand, 5% coarse sand, trace fine gravel, medium dense, wet, gradually fining downward		
						SP: 10YR5/3 WELL SORTED FINE SAND, non plastic, sub rounded to sub angular sand grains and gravel, trace silt, 80% fine sand, 20% medium sand, trace coarse sand, trace fine gravel, medium dense, wet		
						No Recovery		
						SP: 10YR5/3 WELL SORTED FINE SAND, non plastic, sub rounded to sub angular sand grains, trace silt, 95% fine sand, 5% medium sand, trace coarse sand, massive, medium dense, wet, sharp boundary below		
						SW: 10YR5/3 POORLY SORTED MEDIUM SAND, non plastic, sub rounded to sub angular sand grains and gravel, trace silt, 15% fine sand, 60% medium sand, 20% coarse sand, 5% fine gravel, trace coarse gravel, medium dense, wet		
						No Recovery		

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FIELD BOREHOLE LOG						BOREHOLE NO:	120TH-B001	
						TOTAL DEPTH:	50 FT	
DEPTH	SAMPLE TYPE	ATTEMPT	RECOVERY	SOIL SYMBOLS	USCS	SOIL DESCRIPTION	WELL CONSTRUCTION	SAMPLE
35	DIRECT PUSH							
				SP	SP: 10YR5/3 WELL SORTED FINE SAND, non plastic, sub rounded to sub angular sand grains, trace silt, 95% fine sand, 5% medium sand, trace coarse sand, medium dense, wet, coarsening downward	35		
				SP	SP: 10YR5/3 WELL SORTED FINE SAND, non plastic, sub rounded to sub angular sand grains and gravel, trace silt, 85% fine sand, 10% medium sand, 5% coarse sand, trace fine gravel, medium dense, wet, coarsening downward	36		
				SW	SW: 10YR5/3 POORLY SORTED FINE SAND, non plastic, sub rounded to sub angular sand grains and gravel, trace silt, 65% fine sand, 20% medium sand, 10% coarse sand, 5% fine gravel, trace coarse gravel, medium dense, wet, coarsening downward	37		
	DIRECT PUSH			SW	SW: 10YR5/3 POORLY SORTED FINE SAND, non plastic, sub rounded to sub angular sand grains and gravel, trace silt, 40% fine sand, 30% medium sand, 15% coarse sand, 10% fine gravel, 5% coarse gravel, medium dense, wet		38	
					No Recovery		39	
40				SP	SP: 10YR5/3 WELL SORTED FINE SAND, non plastic, sub rounded to sub angular sand grains, trace silt, 95% fine sand, 5% medium sand, trace coarse sand, medium dense, wet, coarsening downward	40		
				SW	SW: 10YR5/3 POORLY SORTED FINE SAND, non plastic, sub rounded to sub angular sand grains and gravel, trace silt, 70% fine sand, 20% medium sand, 10% coarse sand, trace fine gravel, medium dense, wet, coarsening downward	41		
				SW	SW: 10YR5/3 POORLY SORTED COARSE SAND, non plastic, sub rounded to sub angular sand grains and gravel, trace silt, 35% fine sand, 15% medium sand, 40% coarse sand, 10% fine gravel, trace coarse gravel, gravel average 5mm to 12mm, medium dense, wet	42		
					Lithology not recovered due to VAS Collection	43		
						44	VAS Collection: 120TH-B001-45 GW190225 1510KME	
45				SP	SP: 10YR5/2 WELL SORTED FINE SAND, non plastic, sub rounded to sub angular sand grains and gravel, trace silt, 70% fine sand, 20% medium sand, 10% coarse sand, trace fine gravel, medium dense, wet, coarsening downward	45		
					SW: 10YR5/2 POORLY SORTED FINE SAND, non plastic, sub rounded to sub angular sand grains and gravel, trace silt, 50% fine sand, 30% medium sand, 15% coarse sand, 5% fine gravel, trace coarse gravel, medium dense, wet, coarsening downward	46		

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FIELD BOREHOLE LOG

BOREHOLE NO: **120TH-B001**
TOTAL DEPTH: **50 FT**

DEPTH	SAMPLE TYPE	ATTEMPT	RECOVERY	SOIL SYMBOLS	USCS	SOIL DESCRIPTION	WELL CONSTRUCTION	SAMPLE
50	DIRECT PUSH				SW			
					SW	SW: 10YR5/2 POORLY SORTED MEDIUM TO COARSE SAND, non plastic, sub rounded to sub angular sand grains, trace silt, 20% fine sand, 30% medium sand, 30% coarse sand, 15% fine gravel, 5% coarse gravel, medium dense, wet		
					CH	CH: 10YR5/1 WELL SORTED FAT CLAY, high plasticity, sub rounded to sub angular sand grains, 80% clay, 15% silt, 5% fine sand, very stiff (penetrometer 2.5tsf), moist		